

Plumb — project context

Grip-mounted putting stroke analyzer. A 6-axis IMU in the butt of a putter grip derives face angle, tempo and path, and displays them on a 240×240 round LCD. Standalone: no app, no network.

Source of truth:

[docs/superpowers/specs/2026-09-15-putting-analyzer-design.md](#). Read it before writing code. Section numbers below refer to it. If this file and the spec ever disagree, the spec wins — and say so rather than silently picking one.

This is a portfolio project. Where a faster path and a more defensible path conflict, take the defensible one. A finished device with unvalidated numbers fails the goal.

Hard invariants

These are the decisions that fail *silently* — code that looks correct, runs fine, and produces wrong numbers. Do not change any of them without explicit approval from Will.

1. **No putter-type priors anywhere in the algorithm.** Do not normalize face rotation against expected magnitude. Do not tune segmentation thresholds on rotation amplitude. A zero-torque putter legitimately produces very little face rotation; an algorithm that assumes "normal" rotation will report it as broken. Thresholds key off timing and acceleration, never rotation size. (§2.1)
2. **Accelerometer correction gain goes to ~0 during the stroke.** Between BACKSWING and FOLLOWTHROUGH the accelerometer reads gravity *plus stroke acceleration*. A stock Madgwick/Mahony filter will be dragged off attitude by the motion it is supposed to be measuring. Restore the gain in IDLE and ADDRESS. (§7.2)
3. **Face angle is relative to address, not to the target line.** There is no external heading reference and a magnetometer is unusable next to a steel putter head. Never label the output as absolute face angle. (§3.3)
4. **Gyro bias is re-nulled every stroke** during the ADDRESS stillness window. Drift is bounded by the short (~1.5 s) integration window, not by accelerometer correction. This is not optional. (§7.2, §8.5)
5. **Detection thresholds are derived from logged data, never guessed.** They get fixed in the Python notebook during Phase 2 and only then ported. If you need a threshold

before real data exists, parameterize it and leave it configurable — do not hardcode a plausible-looking number. (§7.1)

6. **Wi-Fi stays compiled out in every build.** One firmware configuration, and it is the one that gets validated. The Wi-Fi stack changes interrupt behavior and therefore sampling jitter, which is what the whole accuracy argument rests on. No OTA. (§14.4)
 7. **LVGL draw buffers live in internal SRAM, not PSRAM.** This board is the ESP32-S3R2 — 2 MB *quad* PSRAM, roughly half the bandwidth of the octal parts used in published benchmarks. PSRAM is for fonts and image assets only. (§6.3)
 8. **Sensor acquisition is pinned to core 0, LVGL to core 1,** and nothing renders between arm and follow-through. (§6.2)
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Stack

- **Firmware:** ESP-IDF + FreeRTOS, C. Target [esp32s3](#).
- **UI:** LVGL **9.2 or later** (9.0 carried a ~37% ESP32-S3 regression). UI layer must be pure LVGL with zero ESP-IDF calls so it builds against the PC simulator.
- **Analysis:** Python 3 + NumPy + SciPy, Jupyter notebooks under [analysis/](#).
- **Board config traps:** PSRAM mode **QSPI** (not OPI); flash size **16 MB**; in Arduino IDE only, "USB CDC On Boot" must be **Disabled**. (§14.2)

Layout

/firmware	ESP-IDF project
/analysis	Python: synthetic harness, algorithm development, notebooks
/data	Logged stroke corpus (raw int16 + header)
/hardware	Printed base/puck models, calibration jig, tap-test fixture
/docs	Spec, validation report, accuracy study

Conventions

- **Test-driven where testable.** The synthetic stroke harness ([analysis/](#)) is the algorithm's test suite — ground truth is known because it was generated. Write the failing test first.
- Algorithms are developed and proven in Python, then ported to C and **re-verified against the same corpus**. Do not develop numerical methods directly in firmware; there is no JTAG on this board (§14.3).
- Store raw samples, never derived metrics. The corpus must stay usable when the algorithm changes. (§11)

- Prefer small, single-purpose modules. The UI module takes a results struct and knows nothing about hardware.

Do not

- Do not invent accuracy numbers, or describe anything as validated that has not been measured.
- Do not add features outside §1.2 (no app, no cloud, no Wi-Fi, no green reading, no full-swing support).
- Do not write firmware that assumes hardware is present — it is in transit. See [HANDOFF.md](#) for what is unblocked.